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CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 02 卷 第 07 册
Structure SPECIALITY VOLUME BOOK

卷册 名称 #2继电器室及计量小室结构图
Volume name Structure drawings for No.2 relay room and measuring room

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图纸 6 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

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注:

0 本工程为一层框架结构。

1 本工程结构设计采用的主要楼层面均布活荷载标准值见下表：

荷载类别	标准值(kN/m²)	荷载类别	标准值(kN/m²)
走廊	2.5	不上人屋面	0.7
蓄电池室	20	屋面	1.0
设备机房	按具体设备定		

1 The standard values of live load on the main floor and roof used in the structural design of this project are shown in the following table：

Occupancy or Use	kN/m²	Occupancy or Use	kN/m²
Corridors	2.5	Non-accessible roofs	0.7
storage battery	20	Canopies	1.0
Equipment rooms	Determined by specific equipment		

2 砌体墙体材料的选用见下表：（砌筑砂浆需采用商品砂浆）

部位	厚度(mm)	采用材料	块体强度等级	砂浆强度等级 M表示泥砂浆	最大容重(kN/m³)
0.000以上	230	机制粘土空心砖	f'm=15Mpa(2200psi)	f'g=12.4Mpa(1800psi)	16.4
0.000以下	230	机制粘土空心砖	f'm=23Mpa(3300psi)	f'g=12.4Mpa(1800psi)	16.4

2 The selections of wall materials are shown in the following table:（masonry mortar should be made of commercial mortar）

Members	Thickness(mm)	Materials	Strength grade of masonry units
above 0.000	230	Machine-made	f'm=15Mpa(2200psi)
below 0.000	230	sintered clay brick	f'm=23Mpa(3300psi)

Members	Strength grade of mortar	Maximum unit weight(kN/m³)
above 0.000	f'g=12.4Mpa(1800psi)	16.4
below 0.000	f'g=12.4Mpa(1800psi)	16.4

2.1 内隔墙位置及做法详见建施。

2.1 The location and practice of the internal parting walls are detailed in the building drawings.

2.2 施工图中与手法制图对应的各类构件代号见下表：

构件名称	符号
基础梁、承台	JZL、CT
梁上柱	LZ
普通柱、框架柱、框支柱	Z、KZ、KZZ
非框架梁、框架梁(屋面框架梁)、框支梁	L、KL(WKL)、KZL
悬挑梁、序字梁	XL、JZL
楼板	B
构造柱、过梁、圈梁	GZ、GL、QL
预埋件	M

2.2 The code names in working drawings corresponding to the standard drawings are shown in the following table：

Members	Symbols
Foundation beams, foundation slab	JZL、CT
Columns on the beam	LZ
Ordinary columns,frame columns,frame-supporting columns	Z、KZ、KZZ
Non-framework beams,frame beams(roof frame beams),frame-supported beams	L、KL(WKL)、KZL
Suspension beams,cross ribbed beams	XL、JZL
Floor slabs	B
Constructional columns,over beams,ring beams	GZ、GL、QL
Embedded objects	M

3. 构造要求

3. Detailing requirements

3.1 钢筋的混凝土保护层厚度（最外层钢筋外边缘至混凝土表面的距离）：

混凝土暴露等级	构件	钢筋	保护层厚度mm
混凝土处在地基上并未浇灌在地基土环境中	基础	所有	76.2
	梁 柱	所有	50.8
混凝土处在地基土环境或露天环境中	所有	Φ19~Φ57	50.8
		小于等于Φ16	38.1
混凝土不处在地基土环境或露天环境中	板、墙、容器壳结构	Φ43、Φ57	38.1
		小于等于Φ36	19.1
	梁 柱	主筋、柱箍筋、梁箍筋、螺旋箍筋	38.1

3.1 The thickness of the protective layer of the reinforcement (the distance from the outer edge of the outermost reinforcement to the surface of the concrete)：

concrete exposure	member	reinforcement	cover mm
cast against and permanently in contact with ground	foundation	all	76.2
	beams, columns	all	50.8
exposed to weather or in contact with ground	all	Φ19~Φ57	50.8
		Φ16 and smaller	38.1
not exposed to weather or in contact with ground	slabs,jists,and walls	Φ43、Φ57	38.1
		Φ36 and smaller	19.1
	beams, columns	primary reinforcement, stirrups,ties,spirals,and loops	38.1

3.2 当梁、柱、墙中纵筋的保护层厚度不小于40mm时，应在保护层内设置Φ10@150钢筋网片或采取其他有效的防裂构造措施，网片钢筋的保护层厚度不应小于25mm。

3.2 When the thickness of protective layer of longitudinal bar in beams,columns and walls is not less than 40mm,Φ10@150 steel mesh should be set up in the protective layer or take other effective measures to control the crack.The thickness of protective layer of the mesh bar should not be less than 25mm.

4.1 受力钢筋的锚固。

4.1 The anchorage of the stressed bar.

4.1.1 纵向受拉钢筋的最小锚固长度l_a。

钢筋种类	抗震等级	钢筋直径	混凝土强度(C) 21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

4.1.1 The minimum anchorage length of longitudinal tensile reinforcement L_a and seismic anchorage length.

Type of bar	Class of seismic measure	Diameter of bar	Concrete strength fc' 21.0	28.0
		10	53d	52d
		13	53d	45d
		16	53d	45d
		19	53d	45d
		22	52d	45d
		25	52d	45d

4.1.2 任何情况下，纵向受拉钢筋的锚固长度不应小于250mm。

4.1.2 In any case, the anchorage length of the longitudinal tensile bar shall not be less than 250mm .

4.1.3 当钢筋在混凝土施工过程中易受扰动时，表中锚固长度应乘以修正系数1.1。

4.1.3 When the steel bar is susceptible to disturbance during the construction of concrete, the anchor length in the table shall be multiplied by the correction factor 1.1.

4.1.4 纵向受拉钢筋末端采用弯钩或机械锚固措施时，包括弯钩或锚固端头在内的锚固长度（投影长度）可取表中数字的0.6倍。

4.1.4 When the end of longitudinal tensile bar is bent or mechanically anchored, The anchor length (projection length) including the hook or anchor end,is 0.6 times the figure in the table.

4.1.5 纵向受压钢筋的最小锚固长度不应小于表中数字的0.7倍。

4.1.5 The minimum anchorage length of longitudinally compression bar shall not be less than 0.7 times the figures in the table.

4.2 受力钢筋的连接。

4.2 The connection of the stressed bar.

4.2.1 受力钢筋的连接接头宜设置在受力较小处，在同一根钢筋上宜少设接头。

4.2.1 The connecting joints of the stressed bar should be installed at the less stressed place, and less joints should be installed on the same bar.

4.2.2 钢筋的连接可采用绑扎搭接、机械连接和焊接连接。机械连接接头和焊接连接接头的类型和质量应符合国家现行有关标准的规定。

4.2.2 Strapping lap, mechanical joint and welding connection can be used for connection of the bar.The types and quality of mechanical joint and welding connection shall be in accordance with the current relevant standards of the state.

4.2.3 同一连接区段内受力钢筋的接头面积允许百分率见下表：

接头型式	接头面积允许百分率(%) 受拉区		受压区	连接区段长度
	墙、板、壳	柱		
绑扎搭接接头	≤25,且≤50	≤50	≤50	1.3倍搭接长度
机械连接接头	≤50		不限制	45d(d为纵向受力钢筋的较大直径)
焊接接头	≤50		不限制	45d(d为纵向受力钢筋的较大直径)且不小于500mm

4.2.3 The allowable percentage of joint area of stressed bar in the same connection section is shown in the following table:

Joint types	Allowable percentage (%) Tensile area: Beam,plate,wall	Tensile area: Column	Compressive zone	Length of connection section
Strapping lap	Should be less than or equal to 25,be less than or equal to 50	Should be less than or equal to 50	Should be less than or equal to 50	1.3 times the lap length
Mechanical joint	Should be less than or equal to 50		No limit	45d(d as the larger diameter of longitudinal stressed bar)
Welding joint	Should be less than or equal to 50		No limit	45d(d as the larger diameter of longitudinal stressed bar)≥500mm

4.2.4 纵向受拉钢筋的绑扎搭接长度应根据位于同一连接区段内的钢筋搭接接头面积百分率按下列公式计算：

纵向受拉钢筋搭接长度：$L_l=Q*L_a$

纵向受拉钢筋搭接长度修正系数Q见下表：

纵向受拉钢筋搭接接头面积百分率(%)	<25	50	100
纵向受拉钢筋搭接长度修正系数Q	1.20	1.40	1.60

4.2.4 The lashing lap length of longitudinal tensile bars shall be calculated according to the following formula according to the percentage of overlap area of the bars in the same connecting section:

Overlap length of longitudinal tensile bar: $L_l=Q*L_a$

The coefficient of overlap length of longitudinal tensile bar Q is shown in the following table:

Percentage of lap joint area of longitudinal tensile bar (%)	<25	50	100
Q	1.20	1.40	1.60

4.2.5 任何情况下，纵向受拉钢筋的绑扎搭接接头的搭接长度不应小于300mm。

4.2.5 In any case, the length of the lap joint of the longitudinal tensile bar shall not be less than 300mm.

4.2.6 纵向受压钢筋当采用搭接连接时，其受压搭接长度不应小于纵向受拉钢筋搭接长度的0.7倍，且任何情况下不应小于200mm。

4.2.6 When the longitudinal compressive bar is overlapped, the compressive lap length shall not be less than 0.7 times of the length of the longitudinal tensile bar, and shall not be less than 200mm in any case.

4.2.7 轴心受拉及小偏心受拉杆件（如桁架和拱的下弦）的纵向受力钢筋不得采用绑扎搭接。

4.2.7 Strapping lap shall not be used for the longitudinal stressed bar of axial tension and small eccentric tension members (such as the bottom of truss and arch).

4.2.8 当受拉钢筋直径d>25mm及受压钢筋的直径d>29mm时，不应采用绑扎搭接。

4.2.8 When the diameter of the tensile bar is greater than 25mm and the diameter of the compressed bar is greater than 29mm, the strapping lap shall not be used.

4.2.9 钢筋焊接长度单面10d，双面5d。

4.2.9 The length of welding of steel bar should be 10d in single way, and 5d in both ways.

4.2.10 需要进行疲劳验算的构件，其纵向受拉钢筋不得采用绑扎搭接接头，也不宜采用焊接接头，且严禁在钢筋上焊有任何附件（墙筋锚固除外）。

4.2.10 The longitudinal tensile bar of the members which needs to be checked for fatigue can not be strapping lap joint,and should not be wedding joint.And it is strictly prohibited to weld any attachments on the bar (except for end anchorage).

4.2.11 除满足以上规定外，本工程结构构件受力钢筋采用的连接方式见下表：

构件	抗震等级或部位	受力钢筋连接方式
框架柱	底部	宜采用机械连接 也可采用绑扎搭接连接或焊接连接
墙柱	其他层	可采用绑扎搭接连接或焊接连接
框架梁		宜采用机械连接
		可采用绑扎搭接连接或焊接连接

4.2.11 In addition to the above prescribe, the connection of the stressed bars of the structural members used in this project is shown in the following table:

Members	Segments	Connection of stressed bar
Frame columns	Bottom	Mechanical connection shall be adopted Strapping lap or welding connection can also be used
Boundary columns	Other layers	Strapping lap or welding connection can be used
Frame beams		Mechanical connection shall be adopted
		Strapping lap or welding connection can be used

5.1 避雷接地装法参见电气图纸。

5.1 Methods of lightning protection earthing are shown in electrical drawings.

5.2 沉降观测点做法。

5.2 Methods of settlement observation.

5.2.1 沉降观测工作应在地下室底板完成后，即开始观测，施工期观测可根据施工进度及相关要求确定,竣工后,第一年内每隔2~3个月观测一次,以后每隔4~6个月观测一次。沉降停测标准可采用连续两次半年沉降量不超过2mm。

5.2.1 Settlement observation shall be started after completion of the basement floor, the observation can be determined according to the construction schedule and related requirements. After the final completion, the observation shall be made every 2~3 months for the first year , and shall be observed every 4~6 months afterwards.The settlement test standard can be that the settlement within six months shall not exceed 2mm for two consecutive times.

5.2.2 根据建筑物的重要性、使用要求、基础类型、工程地质条件及预估建筑物沉降等因素综合确定本工程沉降水准测量等级为二等,其闭合差为±0.5√N（N为测站数）。

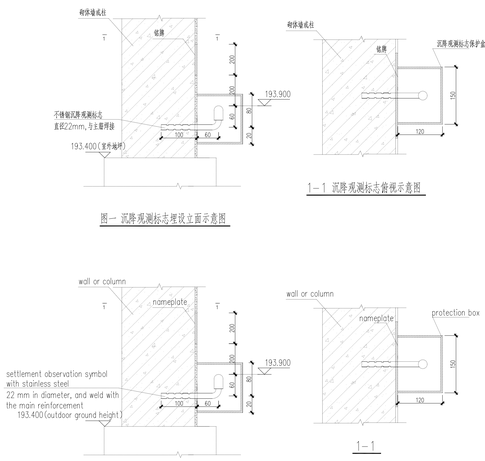
5.2.2 According to the importance of the building, the requirement of use, the type of foundation, the condition of engineering geology and the prediction of the settlement of the building, the grade of the settlement leveling of the project is determined to be two, the closure error is ±0.5√N(N as the number of stations).

5.2.3 鉴于沉降观测资料连贯性的要求,严禁任意改用水准点和更换其标高。

5.2.3 In view of the consistency of settlement observation data, it is strictly forbidden to change the bench mark and the elevation.

5.2.4 ▲表示沉降观测点的平面位置，均设在柱或剪力墙上。根据观测要求的不同采用不同的沉降观测点，见图一。

5.2.4 ▲represents the plane position of the settlement observation points, both on the column or shear wall. According to the different observation requirements, different settlement observation points are shown in Figure 1.



图一 沉降观测标志埋设立面示意图

1—1 沉降观测标志俯视示意图

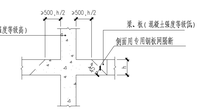
Figure 1 The embeddind method of settlement observation signal

5.3 不同混凝土强度等级的梁柱节点的做法。

5.3 Methods of beam-column nodes with different strength grades of concrete.

梁柱节点处的混凝土应按柱子混凝土强度等级单独浇筑（见图二），在柱混凝土初凝前即浇筑梁板混凝土，并加强混凝土的振捣和养护。

The concrete at the beam-column nodes shall be poured separately according to the strength grade of the column (see Fig.2).Before the initial set of the concrete of column, concrete should be poured into the beam and slab, and the vibration and maintenance of the concrete shall be strengthened.



图二 梁柱节点区不同混凝土强度等级时的做法

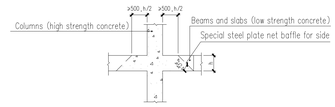


Figure 2 Methods of beam-column nodes in different strength grades of concrete

5.4 混凝土墙、板及梁的洞口加固措施（除注明外）。

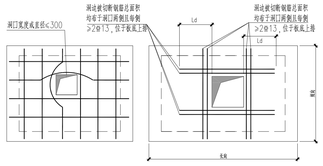
5.4 Measures for reinforcement of openings in concrete walls, slabs and beams (except as noted).

5.4.1 未经结构同意，不得任意增加留洞。

5.4.1 No additional holes may be added without professional consent of the structure.

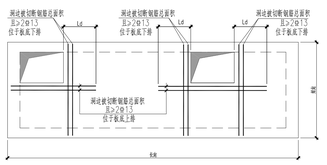
5.4.2 楼板预留孔洞的加固（见图三及五）。

楼板开洞，当洞口直径或宽度≤300时，可不设附加钢筋;板内钢筋不得切断，应沿洞边绕过（见图三）;当洞口直径或宽度>300且≤1000,且洞过不设梁时，加固方式如图四及五所示（此方式不适用于无梁楼盖）;当洞口直径或宽度>300,且在洞口周边有集中荷载，及洞口直径或宽度>1000时,洞边应设过梁,加固方式详见结施。



图三 楼板孔洞加固示意图1

图四 楼板孔洞加固示意图2



图五 楼板孔洞加固示意图3

5.4.2 Reinforcement of preformed holes in floor slabs(seeFig.3and Fig.5).

When the diameter or width of the hole is less than 300,additional bars can not be set,and the bars in the slab shall not be cut off and shall be bypassed along the edge of the hole(see Fig.3).When the diameter or width of the hole is more than 300 and less than or equal to 1000 without beams in the edge of the hole, the reinforcement methods are shown in figure 4 and figure 5（does not apply to the beamless floor).When the opening diameter or width is more than 300, and there is a concentrated load around the entrance, and the diameter or width of the hole is more than 1000, the edge of the hole shall be provided with side beams.Refer to structural drawings for details of methods of reinforcement.

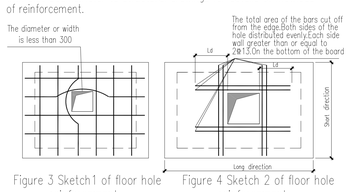


Figure 3 Sketch1 of floor hole reinforcement

Figure 4 Sketch 2 of floor hole reinforcement

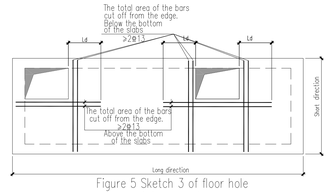
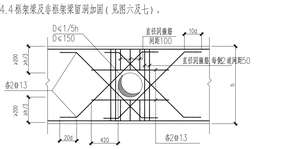


Figure 5 Sketch 3 of floor hole reinforcement

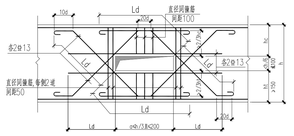
5.4.3 楼板内预埋管的直径不得大于1"(25.4mm)，并以一排为原则，交叉布线须采用线盒，不得立体交叉穿越。板内预埋管的混凝土保护层厚度不得小于38mm。

5.4.3 The diameter of the embedded pipe in the floor shall not be greater than 1"(25.4mm), With the principle of a row, cross wiring must use line box, not cross crossing.The thickness of the concrete protective layer in the pre embedded pipe shall not be less than 38mm.

5.4.4 框架梁及非框架梁留洞加固（见图六及七）。



图六 梁开小洞加固示意



图七 梁开较大洞加固示意

5.4.4 Reinforcement of preformed holes in beams(see Figure.6 and Figure.7).

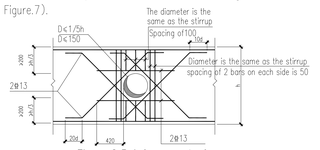


Figure 6 Reinforcement of small opening of beams

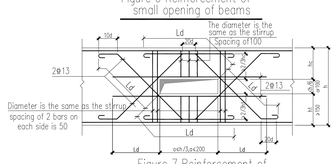


Figure 7 Reinforcement of large opening of beams

5.4.5 楼梁墙洞口宽度小于300mm的预留洞，应以结施图所示为准其他专业图纸或设计修改通知与本条说明有矛盾时，应征得结构设计人员同意并采取有效的技术措施后,方可施工；

5.4.5 The reserved hole penetrating through the reinforced concrete slab or load-bearing brick wall with a side length of no less than 300mm, should be based on the structural construction drawings . If any other major drawing or design modification notice is inconsistent with this instruction, the construction can be carried on only before structural designer agrees to the application, and effective technical measures must be taken.

5.4.6 楼板其他构造。

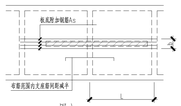
5.4.6 Other measures of slabs.

双向板的短跨主筋应设在长跨主筋的下次；

当楼板上有110及230厚砌体隔墙未设梁而直接支承在板上时，

楼板底筋详图中注明者外应沿砌体墙方向加筋加图八；

The short span main bar of the two-way slab should be located below the long span main bar.When the masonry walls with thickness of 110 and 230 on the floor are not provided with beams and are directly supported on the slab, It shall be reinforced along the masonry wall except as indicated in the detailed drawings at the bottom of slabs(see Figure.8).



图八

跨度L	配筋As	范围b
L≤3000	4Φ13	墙厚+2板厚
3000<L≤3900	4Φ16	

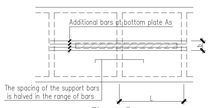


Figure 8

Span L	Area of bars As	Range b
L≤3000	4Φ13	The wall thickness+2
3000<L≤3900	4Φ16	times the plate thickness

6 填充墙的构造等措施及做法。

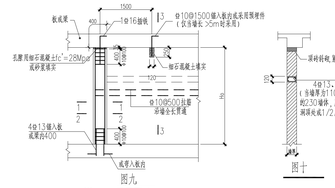
6 filling wall constructional measure and methods of masonry walls.

6.1 砌体墙构造柱的设置位置、断面尺寸见各层建筑平面及相关结施图纸。构造柱与墙连接处砌成马牙槎，必须先砌墙后浇筑。构造柱连接的墙垛长度尺寸按小砌块困难时，可将墙垛改用混凝土，并与构造柱同时浇筑(图九)。除注明者外，构造柱应在墙体转角、不同厚度墙体交接处、较大洞口两侧（≥1500）、悬臂墙端部布置，构造柱间距≤4米，且不大于层高2倍。断面尺寸可参墙身厚度按图十选用。构造柱应先砌墙后浇筑混凝土，混凝土强度为fc'=28Mpa，

砌墙时，构造柱与墙体连接应加拉筋，构造柱上部与梁板连接处应加箍筋均见图九，砌体墙砌筑砂浆强度不应低于f'g=12.4Mpa(1800psi)。构造柱伸入承台基础，柱上下端1m范围内箍筋间距100mm。

6.1 The location and dimension of the constructional columns of masonry wall shall be shown in the plan of each floor and the relevant drawing.the joints of the constructional column and the wall are made horse teeth. The walls must be made before pouring. When the length of the wall connected with the constructional column is small, and which is difficult for construct, the wall can be changed into concrete, and can be poured simultaneously with the constructional column.(see figure 9)

In addition to the relevant drawings indicated, constructional columns should be arranged in the corner of the wall,the junction of walls of different thickness the two sides of the large holes(more than 1500) and the end of cantilever walls.The spacing of constructional columns is less than 4 meters, and not more than 2 times the floor height.The section size can be selected according to the thickness of the wall in Figure.9.Walls should be built before pouring the concrete columns,using concrete of 28Mpa.When building the wall, tie bars should be set in connections of structural columns and walls.The upper part of the structural columns should be reinforced with the connections of beams and slabs, as shown in Figure.9.The strength of masonry mortar shall not be less than f'g=12.4Mpa(1800psi). The constructional column extends into the base of the pile cap. The spacing between stirrups in the 1m range at the upper and lower ends of columns is 100mm.



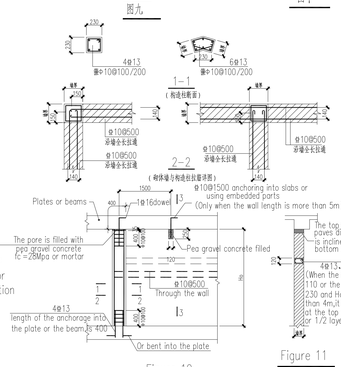
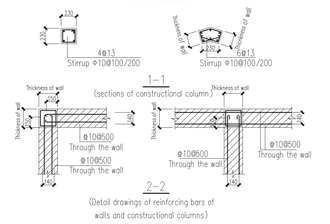


Figure 10

Figure 11



Reference Drawings

Drwg. No.	Title

Revisions	Name	Date	Notes

巴基斯坦默泽亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 0 7 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	郭新勇	
校核 Checked	赵俊	
审核 Reviewed	黄帅	
批准 Approved	胡峰	

#2继电器室及计量小室总说明一

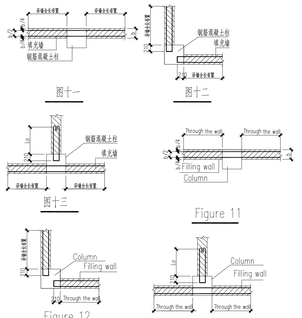
General specification 1 for No.2 relay room and measuring room

比例 Scale | 图号 DWG. NO. 30-B003701S-T0207-001 | 版本 REV. A

B4.TX1189

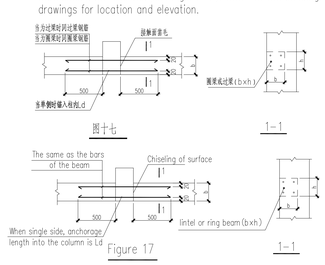
6.2 填充墙与钢筋混凝土柱连接构造。填充墙(110、230 厚)与框架柱连接处设拉接筋详见图十一至图十三，浇筑框架柱时，应配合建施有关图纸按图中要求位置预留φ10@500拉筋。拉接筋应为沿墙全长拉通。楼梯间和人流通道的填充墙在通道向外侧侧，尚应采用钢丝网砂浆面层加强。楼梯间尚应设置间距不大于层高且不大于4米的钢筋混凝土构造柱。

6.2 Connections of filling wall and reinforced concrete column. The connections between the infill wall and the frame column are provided with tie bars, as shown in figures 11 to 13, When pouring a frame column,φ10@500 should be reserved according to the requirement in the picture position cooperating with the relevant drawings.The ties bars should be along the full length of the wall to pull through.The filling walls of staircases and channels (in both sides) should be strengthened with wire mesh mortar surface.In the staircases,constructional columns with a spacing of not more than 4 meters shall be provided.



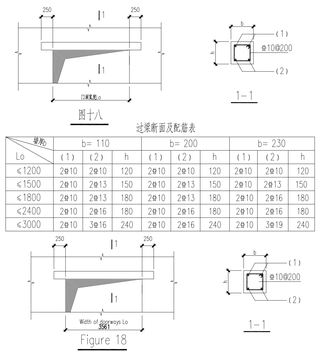
6.3 过梁、圈梁与钢筋混凝土柱连接构造。
凡过梁或圈梁当一端支承在柱时，应预留拉筋，见图十七，位置及标高参见建施有关图纸。

6.3 Connections of lintel, ring beam and column.
When the end is supported in the column,the dowel bars should be reserved(see Figure 17). Refer to the building drawings for location and elevation.



6.4 砌体墙中的门，窗洞及设备预留孔洞洞顶需设过梁；除已有详图注明外断面及配筋按图十八及过梁断面及配筋表选用，混凝土强度为fc'=28Mpa，位置及过梁底标高见有关建施图纸，表中过梁荷重仅考虑1/3Lo高度墙体自重，当超过或梁上作用有其他荷载时另详。

6.4 lintels should be set up on the top of reserved holes of the door,window and equipment holes in masonry wall. In addition to the existing detail drawings stated,the section and bars of lintels should be selected according to Figure 18 and table of sections and bars. The concrete of lintels is 28Mpa. The locations and the bottom elevations of lintels are referred to the building drawings.The load on the lintel is only the dead weight of the wall height of 1/3Lo. Attaching detail drawings in other cases.



过梁断面及配筋表

墙厚b	b= 110			b= 200			b= 230		
Lo	(1)	(2)	h	(1)	(2)	h	(1)	(2)	h
≤1200	2φ10	2φ10	120	2φ10	2φ10	120	2φ10	2φ10	120
≤1500	2φ10	2φ10	150	2φ10	2φ13	150	2φ10	2φ13	150
≤1800	2φ10	2φ13	180	2φ10	2φ13	180	2φ10	2φ16	180
≤2400	2φ10	2φ16	180	2φ10	2φ16	180	2φ10	2φ16	180
≤3000	2φ10	3φ16	240	2φ10	2φ16	240	2φ10	3φ19	240

Table of sections and bars of the lintels

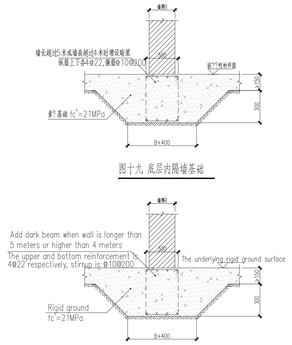
墙厚b	b= 110			b= 200			b= 230		
Lo	(1)	(2)	h	(1)	(2)	h	(1)	(2)	h
≤1200	2φ10	2φ10	120	2φ10	2φ10	120	2φ10	2φ10	120
≤1500	2φ10	2φ13	150	2φ10	2φ13	150	2φ10	2φ13	150
≤1800	2φ10	2φ13	180	2φ10	2φ13	180	2φ10	2φ16	180
≤2400	2φ10	2φ16	180	2φ10	2φ16	180	2φ10	2φ16	180
≤3000	2φ10	3φ16	240	2φ10	2φ16	240	2φ10	3φ19	240

6.5 当后砌非承重墙洞边为混凝土柱时，须在过梁标高处的柱内预埋过梁钢筋，待施工过梁钢筋时，将过梁底筋及架立筋与之焊接；

6.5 When the side of the non bearing wall hole is the concrete column, bars must be embedded in the column at the lintel elevation.In the construction of lintel bars , the bottom and vertical bars should be welded with the lintel bars.

6.6 非承重墙下基础大样见图十九。

6.6 The base of the non-bearing wall is shown in figure 19.

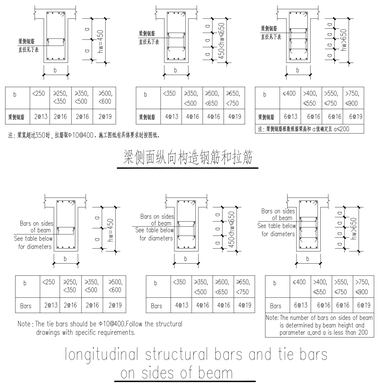


7.其他构造

7. Other detailing requirements

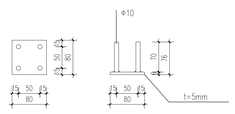
7.1 梁侧面纵向构造钢筋和拉筋详图

7.1 Detail drawings of longitudinal structural bars and tie bars on sides of beam



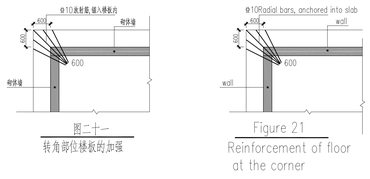
7.2 栏杆百叶预埋件，具体位置详建筑。

7.2 Built in fitting of handrail see details below, location see architectural drawings.



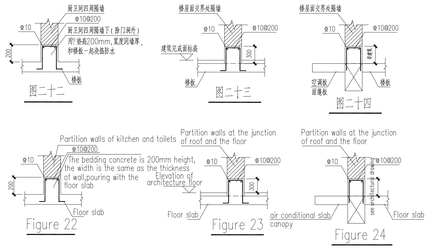
7.3 转角部位楼板的加强排施见图二十一。

7.3 Refer to Figure 21 for the reinforcement of floor at the corner.



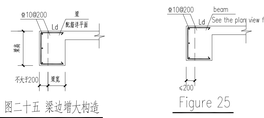
7.4 厨房、卫生间与其它室内房间相邻墙体下设置混凝土导墙，做法见图二十二。
楼面面层防水止水构造，屋面与相邻墙交界处止水构造做法见图二十三。
空调板、雨篷板后砌口见图二十四。

7.4 The concrete walls are set up under the walls of the kitchen, toilet and other indoor rooms, as shown in Figure 22. Water seal structure at the junction of roof and the floor and the junction of roof and masonry wall are shown in figure 23. The partition walls beside air conditional slab or canopy are shown in figure 24.



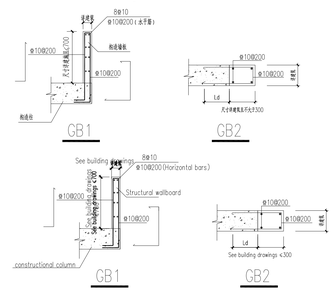
7.5 因建筑需要，梁边增大部分构造见图二十五。

7.5 Refer to Figure 25 for the enlarged side of the beam.



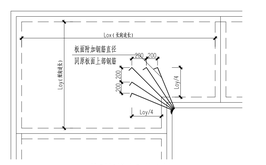
7.6 阳台板、空调板、窗台板上竖向构造墙板配筋及与主体连接见下图。构造墙板后做，主体结构施工时预留拉筋。

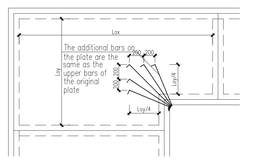
7.6 The bars of the vertical structure panel on the balcony board, air conditioning board and window sill board, and the connection with the main part are shown below.the structural wallboard should be set up after the main part,and there shall be embedded bars in the construction stage of main part.



7.7 异形板内折角(包括升降板引起内折角)处钢筋构造要求见下。

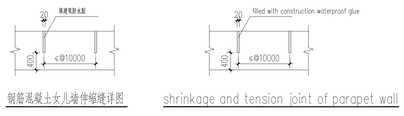
7.7 The structural requirements of the bars in inner fold angle of the sketch plate(including the inner fold angle caused by the lifting plate) are shown below.





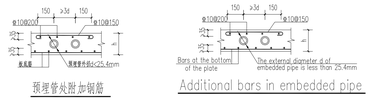
7.8 女儿墙直线长度大于15米时应设伸缩缝。

7.8 When the linear length of parapet wall is more than 15m, shrinkage and tension joint should be set up.

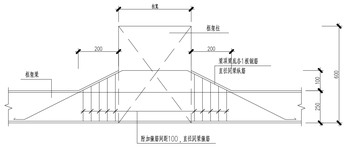


7.9 楼板预埋管处板面加强排施

7.9 Strengthening measures for the slab in embedded pipe.

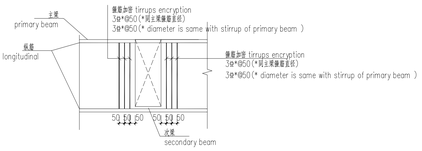


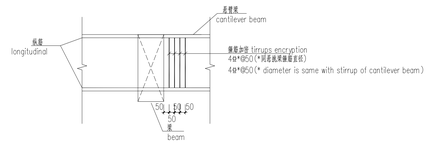
7.10 二层以上框架梁截面中心线与柱中心线偏心距大于 1/4 柱宽时采取框架梁水平加腋排施，加腋厚度同梁高，需水平加腋的梁柱位置见各结构梁配筋平面图。



7.11 框架梁在次梁侧置处左右各附加箍筋3φ*@50(*指梁中箍筋)；次梁在二级次梁侧置处左右各附加箍筋3φ*@50(*指梁中箍筋)，悬臂梁端附加箍筋4φ*@50。

7.11 In the primary and secondary beam intersection node, stirrups are encrypted by 3φ*@50 on both sides of secondary beam (* refers to primary beam stirrup), the cantilever beam root is encrypted by 4φ*@50



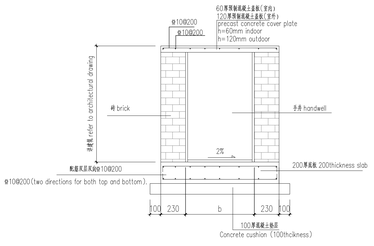


7.12 框架梁在次梁侧置处左右各附加箍筋3φ*@50(*指梁中箍筋)；次梁在二级次梁侧置处左右各附加箍筋3φ*@50(*指梁中箍筋)，悬臂梁端附加箍筋4φ*@50。

7.12 In the primary and secondary beam intersection node, stirrups are encrypted by 3φ*@50 on both sides of secondary beam (* refers to primary beam stirrup), the cantilever beam root is encrypted by 4φ*@50

7.13 建筑手井详见下图，具体尺寸参见建筑图。

7.13 Hand well refer to details below,size according to architectural drawing.



8.平法标注说明：

notes: When a certain value in the concentrated annotation is not applicable to a certain part of the beam, the numerical value is marked in situ. when under construction,in-situ annotation is preferred.

9.钢筋要求 reinforcement requirements:

9.1.框架主体结构抗震钢筋应满足ASTM A 615, Grade 60[420]钢筋要求,本图中以直径表示，具体编号参照下表。

9.1.Seismic reinforcement of frame structure should meet the requirements of ASTM A 615 Grade 60[420] ,reinforcement shows as diameter in the drawing,reinforcement number could refer to table below.

拉伸要求

		Grade 40 [300]A	Grade 60 [420]
最小拉伸强度, psi[Mpa]		70000[500]	90000[620]
最小屈服强度, psi[Mpa]		40000[300]	60000[420]
8 in.[203.2mm]最小伸长率，%			
钢筋名称	3[10]	11	9
	4,5[13,16]	12	9
	6[19]	12	9
	7,8[22,25]	…	8
	9,10,11[29,32,36]	…	7

Grade 40 [300] 钢筋部分提供3 in.(10mm)和in.(19mm)两直径。

Tensile Requirement

	Grade 40 [300]A	Grade 60 [420]
Tensile strength, min, psi[Mpa]	70000[500]	90000[620]
Yield strength, min, psi[Mpa]	40000[300]	60000[420]
Elongation in 8 in. [203.2mm], min, %		
Bar Designation No.3[10]	11	9
4,5[13,16]	12	9
6[19]	12	9
7,8[22,25]	…	8
9,10,11[29,32,36]	…	7

A Grade 40 [300] bars are furnished only in sizes 3 through 6 [10 through 19].

9.2. 框架结构纵向受力钢筋，钢筋的抗拉强度实测值与屈服强度实测值的比值不应小于1.25；钢筋的屈服强度实测值与屈服强度标准值的比值不应大于1.3，且钢筋在最大拉力下的总伸长率实测值不应小于9%。

9.2. For the non-prestersed longitudinal reinforcements of framed structures, the ratio of the actual ultimate tensile strength to actual tensile yield strength shall not be less than 1.25, and the ratio of actual tensile yield sterngth to characteristic strength shall not greater than 1.3, and the total elongation of reinforcement under the maximum tension values should not be less than 9%.

10.参照规范 code:

ASCE 7-10 Minimum Design Loads for Buildings and Other Structures
ACI 530-02 building code requirements for masonry structures
ACI 318-14-2014 building code requirements for structural concrete building code of pakistan
ASTM C 1218/C 1218M-1999 Test Method for Water-Soluble Chloride in Mortar and Concrete
Uniform Building Code 1997
ASTM A 615/A 615M-1 Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE | 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 0 7 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	郑彩娟	
校核 Checked	赵俊	
审核 Reviewed	赵俊	
批准 Approved	郑彩娟	

#2继电器室及计量小室总说明二
General specification 2 for No.2 relay room and measuring room

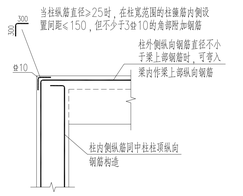
比例 Scale | 图号 DWG. NO. 30-B003701S-T0207-002 | 版本 REV. A

11 抗震边柱和角柱柱顶纵向钢筋构造

11 Detailing of longitudinal bars at the top of edge columns and corner columns.

柱顶纵向钢筋锚固节点详见图三十五~图三十九，图三十八不得单独使用(仅用于未伸入梁内的柱外侧纵筋锚固)，伸入梁内的柱外侧纵筋不宜少于柱外侧全部纵筋面积的65%。

The detailing of longitudinal bars at the joint are shown in the figure 35 ~ figure 39. Figure 38 should not be used alone (only apply to the anchorage of outer layer longitudinal bars in columns which do not extend into beams).No less than 65% of outer layer longitudinal bars in columns should extend into beams.



图三十五

柱筋作为梁上部钢筋使用

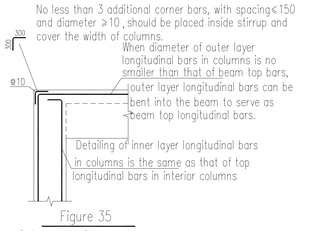
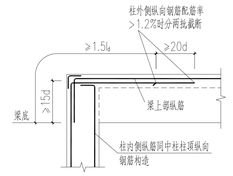


Figure 35

Column reinforcement serves as the top bars of beams



图三十六

从梁底算起 1.5l_a 超过柱内侧边缘

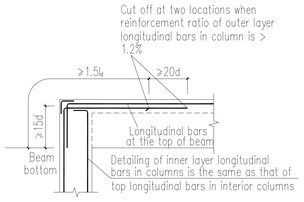
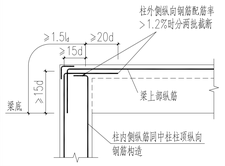


Figure 36

1.5$l_a$measured from the bottom of beam is beyond the inner face of column



图三十七

从梁底算起 1.5l_a 未超过柱内侧边缘

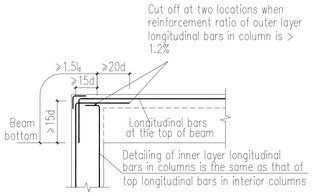
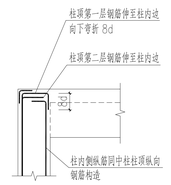


Figure 37

1.5$l_a$measured from the bottom of beam is not beyond the inner face of column



图三十八

(用于图三十六或图三十七节点未伸入梁内的柱外侧钢筋锚固)

当现浇板厚度不小于100时，也可按图三十六节点方式伸入板内锚固，且伸入板内长度不宜小于15d

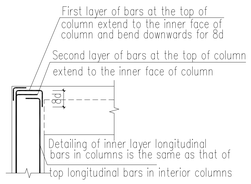
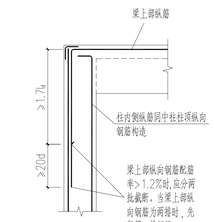


Figure 38

(Apples to the anchorage of outer layer bars beam is not beyond the inner face of column in columns which do not extend into beams as figure 36 and 37.)

When cast-in situ slab thickness is no less than100,bars can be bent into slabs using the same way as figure 36, length of bars extended into slab should not be less than 15d.



图三十九

梁、柱纵向钢筋搭接接头沿节点外侧直线布置

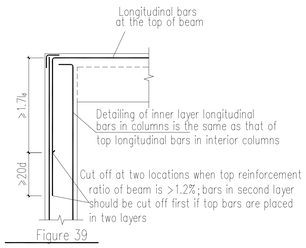


Figure 39

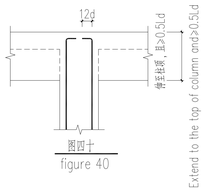
Lap of beam and column longitudinal bars are in a straight line along the outer face of the joint.

节点纵向钢筋弯折要求

Requirements for bending of longitudinal bars at joint

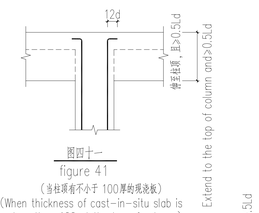
抗震中柱柱顶纵向钢筋构造详见图四十~图四十三

Detailing of top longitudinal bars in interior columns are shown in the figure 40 ~ figure 43.



图四十

figure 40

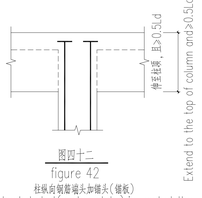


图四十一

figure 41

(当柱顶有不小于 100厚的现浇板)

(When thickness of cast-in-situ slab is no less than 100 at the top of column)

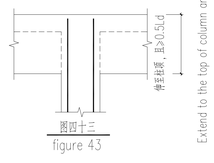


图四十二

figure 42

柱纵向钢筋端头加锚头(锚板)

Headed stud(anchor plate) is used at the end of column longitudinal

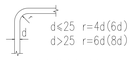


图四十三

figure 43

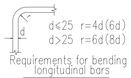
(当直锚长度≥l_a时)

(When straight anchorage length is≥l_a)



纵向钢筋弯折要求

(括号内为顶层边节点要求)

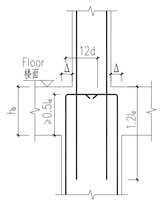


Requirements for bending of longitudinal bars

(requirements of side nodes are in brackets)

柱变截面位置纵向钢筋构造详见图四十四~图四十八

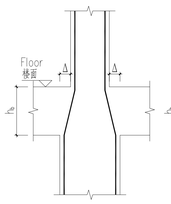
Detailing of longitudinal bars at variable sections of columns are shown in the figure 44 ~ figure 48.



(Δ/h_b>1/6)

图四十四

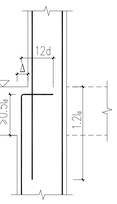
figure 44



(Δ/h_b≤1/6)

图四十五

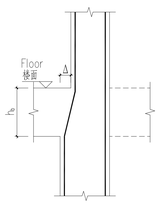
figure 45



(Δ/h_b>1/6)

图四十六

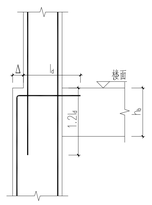
figure 46



(Δ/h_b≤1/6)

图四十七

figure 47



图四十八

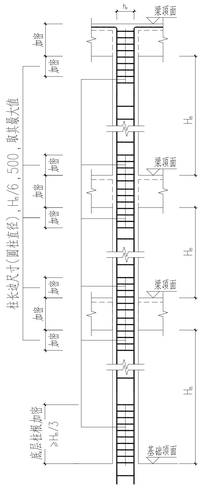
figure 48

12 抗震柱箍筋加密区范围

12 Stirrup densification zone in seismic columns .

当柱在某楼层各向均无梁连接时，计算箍筋加密范围采用的H_n按该联层柱的边净高取用，其余情况同普通柱。

If columns are not connected to any beams or slabs at certain floor,H_n used for calculating the height of stirrup densification zone should be the total clear height of the duplex column.



图四十九

抗震柱箍筋加密区范围

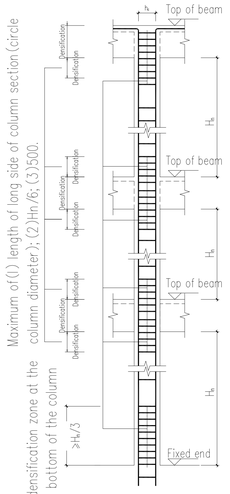
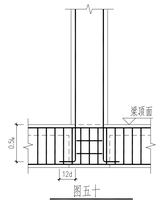


figure 49

Stirrup densification zones of seismic columns



图五十

梁上柱纵筋构造

(在梁内设两道柱箍筋)

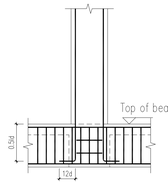


figure 50

Detailing of longitudinal bars in columns supported by beams

(at least two column stirrups should be placed in the beam)

图五十一

底层刚性地面上下各加密

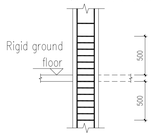
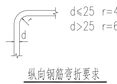


figure 51

Stirrup densification zone for 500 above and below rigid ground



纵向钢筋弯折要求

Requirements for bending of longitudinal bars

13 抗震层面梁纵向钢筋构造

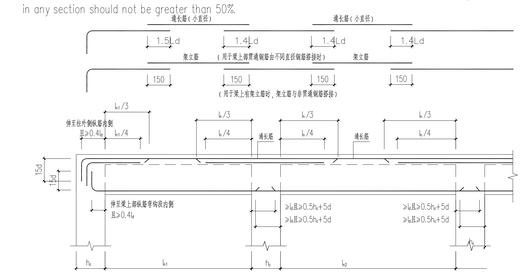
13 Detailing of longitudinal bars in roof beams

13.1 图五十二中，h_c为柱截面沿框架方向的高度，跨度值n为左跨和右跨的较大值。

13.1 In the figure 52, h_c in this drawing is the dimension of column section in the direction of the frame. The span value is the larger value value of the left and right span.

13.2 梁上部通长钢筋与非贯通钢筋直径相同时，连接位置宜位于跨中l_n/3范围内；梁下部钢筋连接位置宜位于支座l_n/3范围内，且在同一连接区段内钢筋接头面积百分率不宜大于50%。

13.2 If the diameters of top continuous bars and discontinuous bars are the same, splices should be within the middle l_n/3 in the span; splices of beam bottom bars should be within l_n/3 of the span around the support;area percentage of spliced bars over total bars in any section should not be greater than 50%.



图五十二

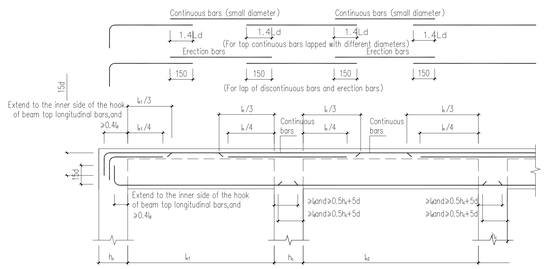
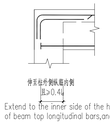


figure 52

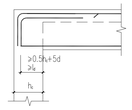


图五十三

figure 53

顶层端节点梁下部钢筋端头加锚头(锚板)锚固

Anchorage of bottom bars in roof beam at end support using headed stud(anchor plate)

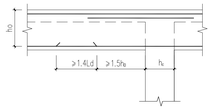


图五十四

figure 54

顶层端支座梁下部钢筋直锚

Straight anchorage of bottom bars in roof beams at end support



图五十五

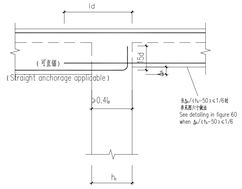
figure 55

顶层中间节点梁下部钢筋在节点外搭接(梁下部钢筋不能在柱内锚固时，可在节点外搭接。相邻跨钢筋直径不同时，搭接位置位于较小直径一跨)

Lap splices of beam bottom bars out of interior joints at intermediate floors (Beam bottom bars can be lapped out of the joint if they can not be anchored in column.When diameters of beam bottom bars are different in adjacent spans, lap splice should be in the span with smaller bar diameter.)

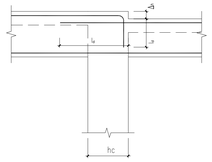
14 梁中间支座纵向钢筋构造

14 Detailing of longitudinal bars in beams at interior supports



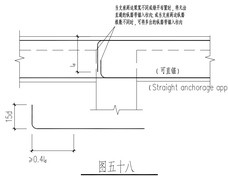
图五十六

figure 56



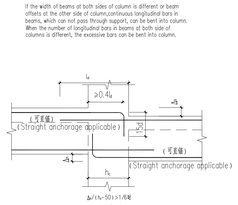
图五十七

figure 57



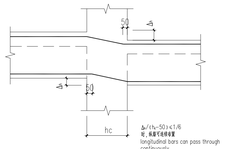
图五十八

figure 58



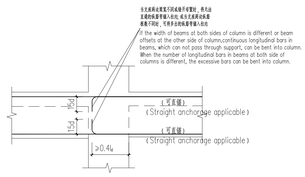
图五十九

figure 59



图六十

figure 60



图六十一

figure 61

Reference Drawings	
Drwg. No.	Title

Revisions			
	Name	Date	Notes

巴基斯坦默摩西里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Mohari to Lahore in Pakistan

CONTRACTOR 国家电网 SGCC 中国电力技术装备有限公司 CHINA TECHNOLOGY EQUIPMENT CO.,LTD.

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 0 7 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	郭承炜	
校核 Checked	赵俊	
审核 Reviewed	陈明	
批准 Approved	王丹	

#2继电器室及计量小室总说明三

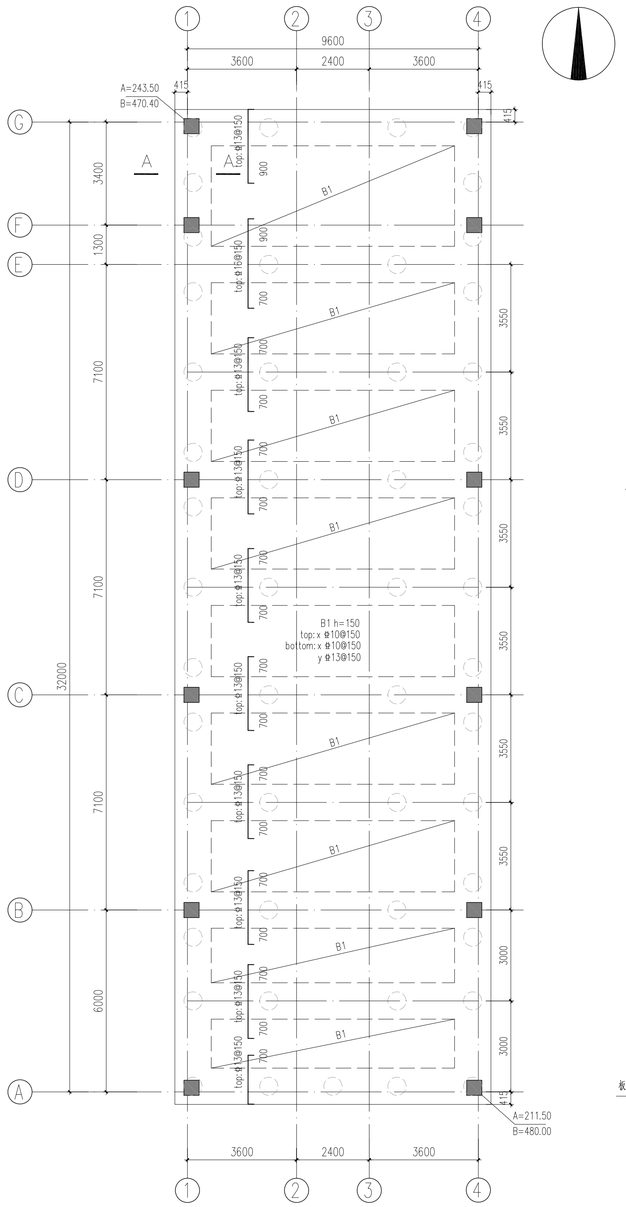
General specification 3 for No.2 relay room and measuring room

比例 Scale

图号 DWG NO. 30-B003701S-T0207-003

版本 REV. A

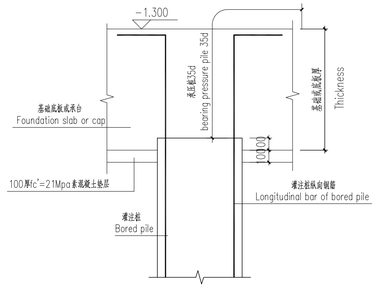
B4TXT169



#2继电器室及计量小室基础施工图

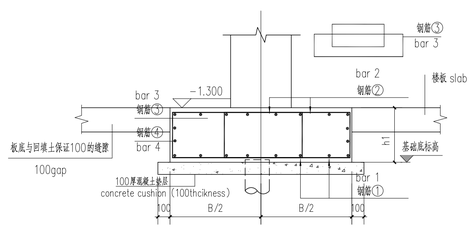
Implementation of mat foundation for No.2 relay room and measuring room

未注明分布钢筋φ10@200



机械成孔灌注桩与底板或基础连接节点图

Joint of the bored pile and bottom plate or foundation



A-A (JC1)

普通基础配筋表：

基础编号	基础平面尺寸 (B)	高度 h1	底部长向钢筋①	顶部长向钢筋②	箍筋③	腰筋④	备注
JC1	1200	600	φ22@150(G60)	φ22@150(G60)	φ10@200(G60)4肢	4φ10(G60)	

foundation reinforcement table：

foundation	size (B)	height h1	bottom long side bar①	upper long side bar①	stirrup③	waist steel④	note
JC1	1200	600	φ22@150(G60)	φ22@150(G60)	φ10@200(G60)4 limb	4φ10(G60)	

说明：

1、±0.000相当于绝对标高193.85(大地水准面EGM2008系统)，室内外高差450mm。
2、本工程基础设计依据《巴基斯坦默蒂亚里±660KV直流输电工程拉合尔±660KV换流站施工图设计阶段岩土工程勘察报告(详勘)》(B003701S-G01-01)进行。
3、承台混凝土强度采用fc'=28Mpa,钢筋ASTM A615 Grade 60, fy = 420MPa。
4、本工程基础采用条形基础+灌注桩的形式，未注明的条形基础均为JC1。
5、除注明外条形基础顶标高为-1.300。
6、本图中未注明梁定位均居轴线中，未注明独立基础定位与柱居中对齐,未注明的柱按轴线居中或与墙平齐。
7.未注明承台及基础梁垫层采用100厚fc'=21Mpa混凝土。
8、地下混凝土要求：
1)水胶比不大于0.50；
2)最大氯离子含量(水泥用量的百分比)：0.10；测试方法参《ASTM C 1218M》。
3)水泥型号：采用抗硫酸盐的硅酸盐水泥，水泥型号需满足《ACI 318-14》表19.3.2.1对应于暴露等级S1的水泥种类要求(ASTM C150 或ASTM 595 或ASTM C1157对应的类别)；
4)水泥中掺入矿物掺合料(矿物掺合料的掺量、使用方法和耐腐蚀性能按相应产品的使用说明并经验证后确定)。
9、建筑完成面到天然地坪之间填土需采用好土回填，并分层夯实，压实系数不小于0.94。
10、本工程初设设计尚未审定，未通过NTDC审核，桩基存在修改的可能，施工前请联系业主与设计确认。
11、首层柱在基础预留插筋长度为1/3Hn，Hn为首层层高。
12、条形基础之间采用现浇混凝土楼板，板厚为150mm，板顶标高为-1.300。
13、隔墙下无梁时，板底附加钢筋参总说明5.4.6,并锚入支座.

Specification：

1. The indoor elevation of ±0.000 is equal to the absolute elevation of193.85 (geodestic datum system EGM2008) If not specified,the elevations of the project are adopt to the relative elevation.The interior and outdoor dispersion is 450mm.
2 .The foundation design of the project proceeds according to《The geotechnical engineering investigation report(Detail survey) of drawing construction design phase of ±660KV converter station in Lahore in Pakistan Matiari ±660KV HVDC project》(B003701S-G01-01).
3.The strength of concrete used in foundation is fc'=28Mpa,reinforcement ASTM A615 Grade 60, fy = 420MPa.
4.The foundation style of this project is strip foundation+cast-in-place piles, the unspecified strip foundation is JC1.
5.The height of the top of the strip foundation is -1.300(if not specified).
6. Foundations are symmetric on the middle of the columns(if not specified),beams are symmetric on the middle of the axis(if not specified), columns are symmetric on the middle of the axis or parallel and level to the edge of the wall(if not specified),
7.Unspecified cusion for foundation and foundation beam is 100 thickness and The strength is fc'=21Mpa.
8. Requirements for Underground Concrete:
1) Maximum water-cement ratio is 0.50;
2) Maximum water-soluble chloride ion content is 0.1;Test method refers to "ASTM C 1218M".
3) Type of Concrete: Concrete should be made of sulfate-resistance silica cement.Cement type should fulfill the requirements for concrete used in exposure grade S1, according to《ACI 318-14》table 19.3.2.1.(ASTM C150,ASTM 595 or ASTM C1157 corresponding cement type).
4) Add mineral admixtures (the dosage, using method and anti-corrosion ability should be tested according to the product manual)
9.The gap between the architectural floor and natural floor should be refilled with good quality soil and tamped by layers, the compacting factor should be more than 0.94.
10. The initial design of the project has not yet been approved by NTDC, the pile foundation is possible to amend.Please contact the owner and designer before construction.
11. The length of joint bar of first floor columns in the foundation should be 1/3 Hn,Hn is the clear height of first floor.
12. Between the strip foundation, the cast-in-situ concrete floor is used, which thickness is 150mm, the height of the top of the floor is -1.300.
13. When there is no beam under partition, additional bars is needed at the bottom of the slab,see general specification 5.4.6.

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND ENGINEERING CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-LCS-ECI-DD-CS-0207-004A

	姓名 Name	日期 Date
设计 Designed	郭新凯	
校核 Checked	赵俊	
审核 Reviewed	张山	
批准 Approved	王海涛	

#2继电器室及计量小室基础施工图

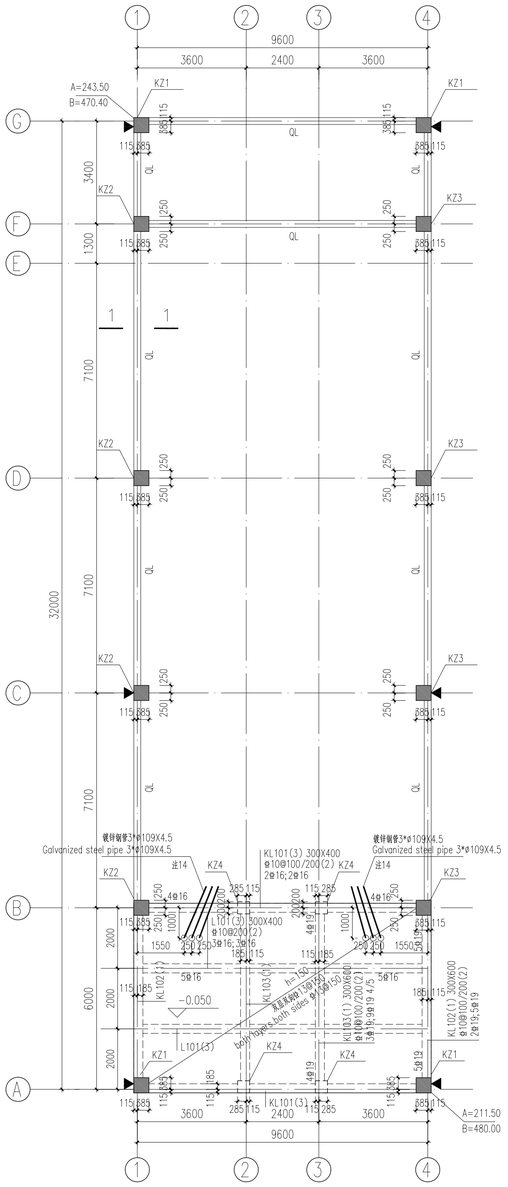
Implementation of mat foundation for No.2 relay room and measuring room

比例 Scale

图号 DWG NO. 30-B003701S-T0207-004

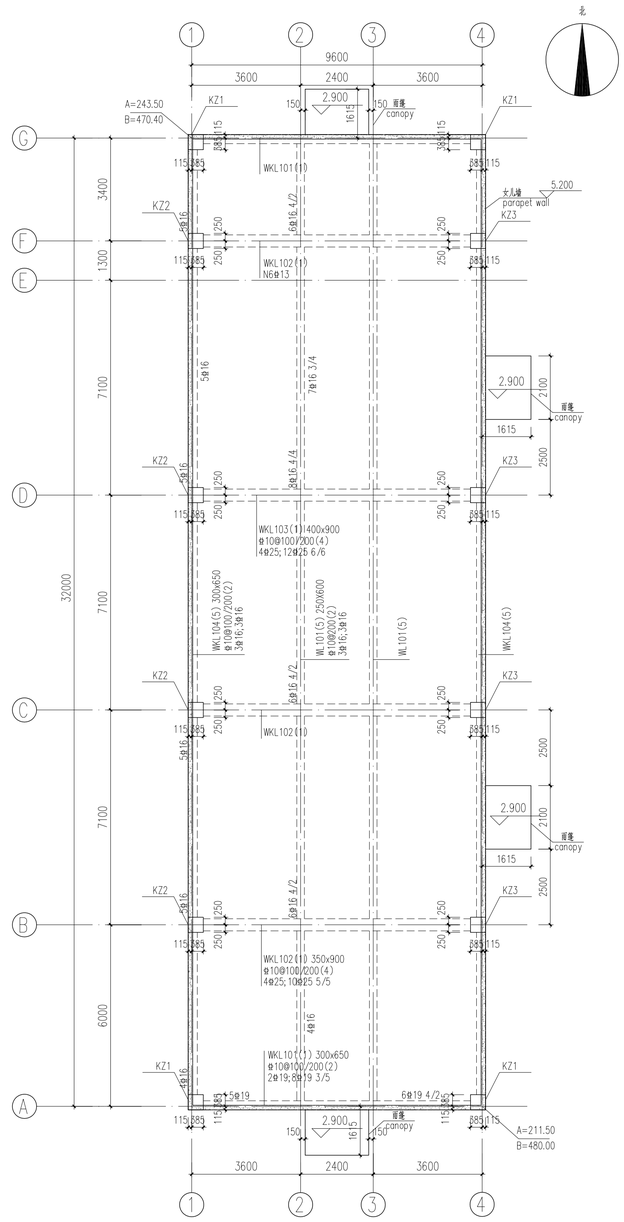
版本 REV. A

594X841



#2继电器室及计量小室-0.050m结构施工图

Implementation of structural members at level -0.050m for No.2 relay room and measuring room



#2继电器室及计量小室4.2m层结构施工图

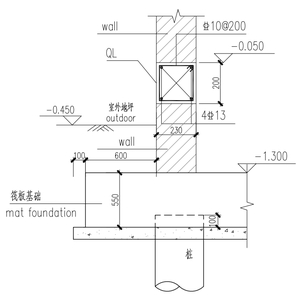
Implementation of structural members at level 4.2m for No.2 relay room and measuring room

截面 section			
编号 No.	KZ1	KZ2 / KZ3	KZ4
标高 height	基础顶 top of foundation~4.200	基础顶 top of foundation~4.200	基础顶 top of foundation~-0.050
纵筋 longitudinal bars	4Φ19(角筋 bar at the corner)	4Φ19(角筋 bar at the corner)	4Φ19(角筋 bar at the corner)
箍筋 stirrups	Φ10@100	Φ10@100(基础顶top of foundation~-0.050) Φ10@100/200 (-0.050~4.200) / Φ10@100	Φ10@100

注：1、图中未注明的屋面现浇板厚为120mm，未注明的板钢筋为双层双向Φ10@100。
2、除特别示明外，图中梁定位未标处，中线与轴线重合；或边线贴墙（柱）边。
3、严禁未经结构同意在梁或柱上留洞。
4、洞口附加钢筋见总说明。
5、混凝土强度为fc'=28Mpa；钢筋ASTM A615 Grade 60，fy = 420MPa；
6、隔墙下无梁时，板底附加钢筋参总说明5.4.6，并锚入支座。
7、施工时应当与建筑及设备各工种施工图核对无误后方可施工。
8、梁上部同直径负筋应尽量拉通。
9、门窗洞口及平面设备洞口定位及大小均详建筑。
10、女儿墙平面位置参建筑图纸。
11、所有混凝土导墙定位及做法详总说明及建筑图纸。
12、▼表示沉降观测点（共6处），具体做法详总说明。
13、本工程初设设计尚未审定，未通过NTDC审核，存在修改的可能，施工前请联系业主与设计确认。
14、每个蓄电池室预埋3根Φ109x4.5的镀锌钢管至继电器室电缆沟内。埋管高出蓄电池室地面300mm。

Specification :

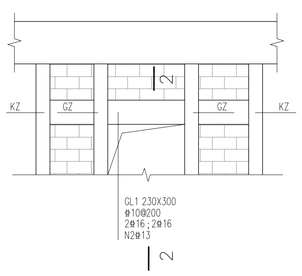
1. The roof slab thickness is 120mm (if not specified), and the reinforcement in the slab is Φ10@100 (both sides and layers)
2. If not specified, all beams are symmetric on the axis or even with the edge of walls and columns.
3. Holes are not allowed on beams and columns unless agreed by structural engineer.
4. Addition reinforcement is needed around the hole , check the specification for details.
5. The material strength of concrete is fc'=28Mpa, reinforcement ASTM A615 Grade 60, fy = 420MPa.
6. When there is no beam under partition, additional bars is needed at the bottom of the slab, see general specification 5.4.6.
7. The drawings should be checked with drawings of other specialty before construction.
8. The upper reinforcement of beams should be continuous if the bar size is same.
9. The location and size of door and window holes and pipeline holes are all detailed in architectural drawings.
10. The position of parapet wall is show in architecture drawings.
11. The position of concrete guide wall is show in architecture drawings, and the practice is shown in general specification.
12. ▼ represents settlement observation points (6 in total), which is shown in general specification.
13. The initial design of the project has not yet been approved by NTDC, the drawing is possible to amend. Please contact the owner and designer before construction.
14. Each battery embeds 3 root Φ109*4.5 of galvanized steel pipe to cable duct of relay room. Embedded pipe is 300 mm above battery room floor.



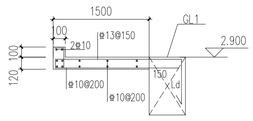
-0.050m圈梁详图

details of ring beam at -0.050m

1-1

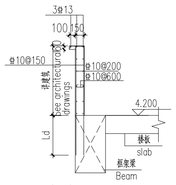


2-2



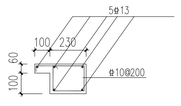
雨篷详图

canopy



女儿墙详图

parapet wall



窗台过梁

window lintel

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 0 7 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	郭彩娟	
校核 Checked	赵俊	
审核 Reviewed	苏刚	
批准 Approved	韩和平	

#2继电器室及计量小室结构施工图

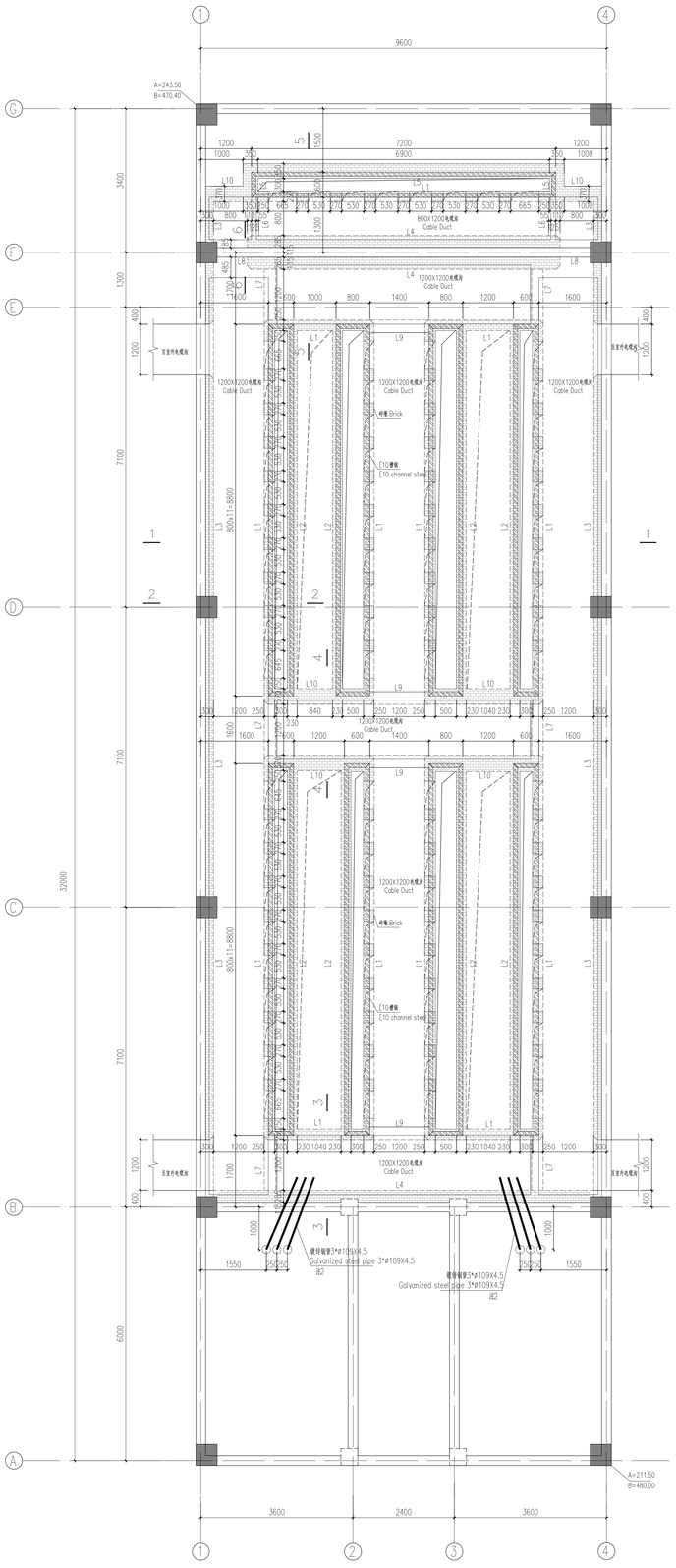
Implementation of structural members for No.2 relay room and measuring room

比例 Scale:

图号 DWG NO.: 30-B003701S-T0207-005

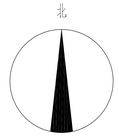
版本 REV.: A

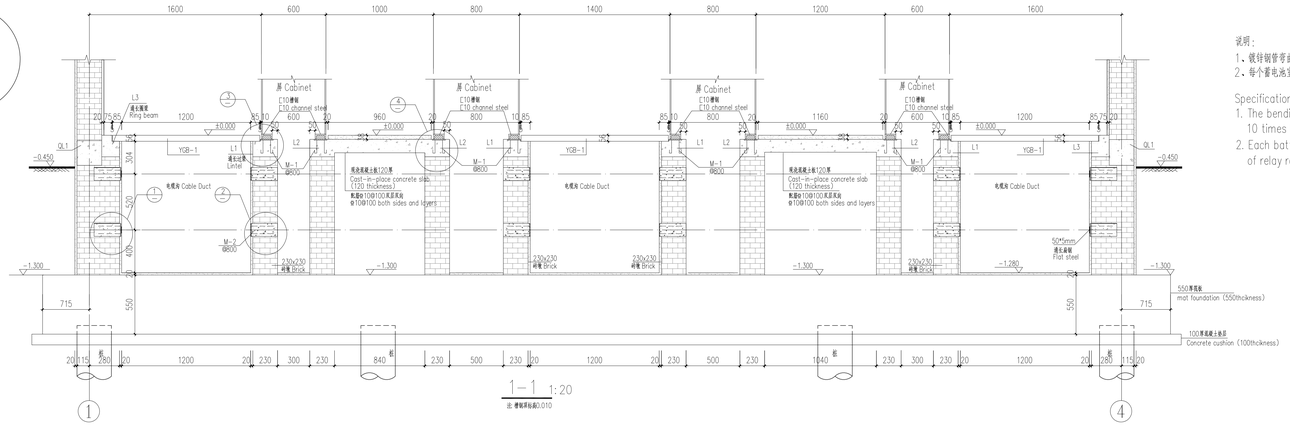
594X841



#2继电器室及计量小室电缆沟平面布置图

Plan layout of cable duct for No.2 relay room and measuring room





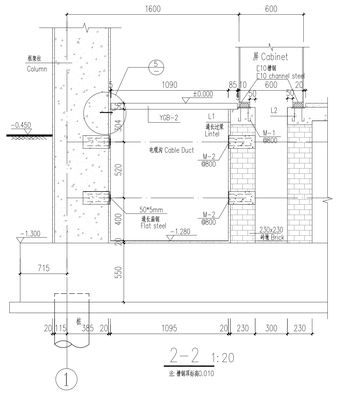
说明：

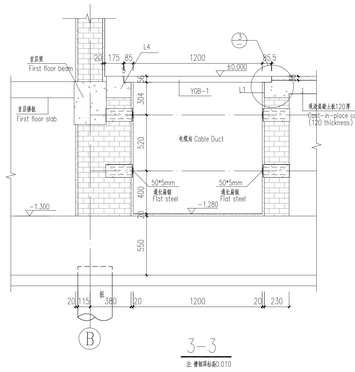
1、镀锌钢管弯曲半径不得小于10倍钢管外径。

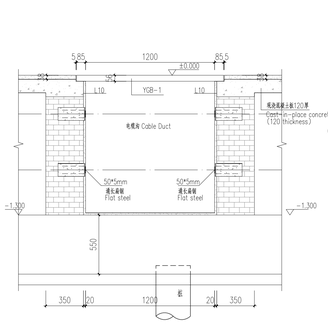
2、每个蓄电池室预埋3根Φ109x4.5的镀锌钢管至继电器室电缆沟内。埋管高出蓄电池室地面300mm。

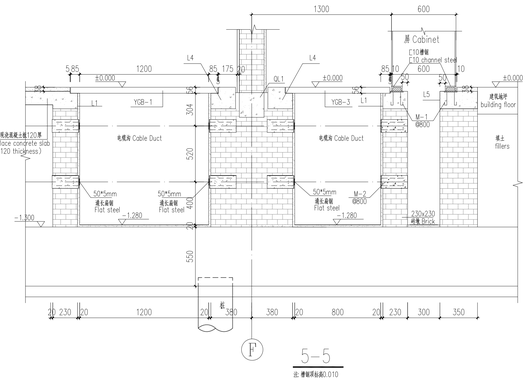
Specification :

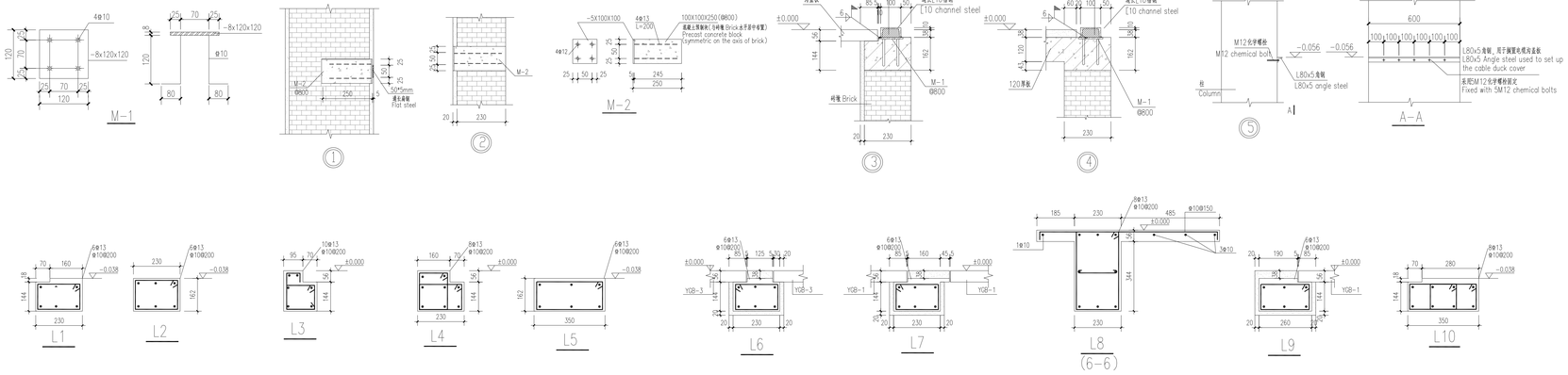
1. The bending radius of the galvanized steel tube shall not be less than 10 times the diameter of the steel pipe.
2. Each battery embeds 3 root Φ109x4.5 of galvanized steel pipe to cable duct of relay room. Embedded pipe is 300 mm above battery room floor.

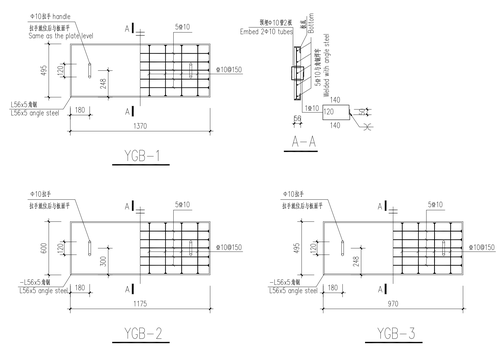


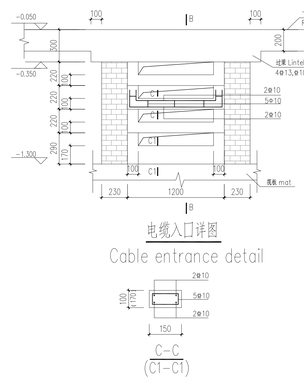


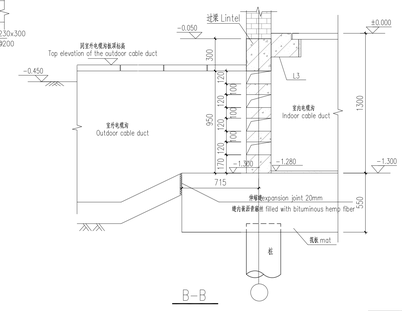












Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

CONTRACTOR: 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA EQUIPMENT

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

SUBCONTRACTOR: East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE | 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 0 7 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	郭新湧	
校核 Checked	赵俊	
审核 Reviewed	陈鹏	
批准 Approved	王辉	

#2继电器室及计量小室电缆沟平面布置图

Plan layout of cable duct for No.2 relay room and measuring room

电气		
专业 Speciality	会签 Sign	日期 Date

比例 Scale | 图号 DWG. NO. 30-B003701S-T0207-006 | 版本 REV. A